

HS 491-VIII Challenges in the Digital Society

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The course addresses societal changes and risks created by the digital spaces we live in today. It provides a social and technological perspective on key topics like ethics and cultural perspectives - privacy and trust including historical background, surveillance, privacy technologies, cryptography basics – algorithms, bias, and transparency, regulation and law related with digital technologies.

The classes will take place (depending on the calendar of my visit to the Institute)

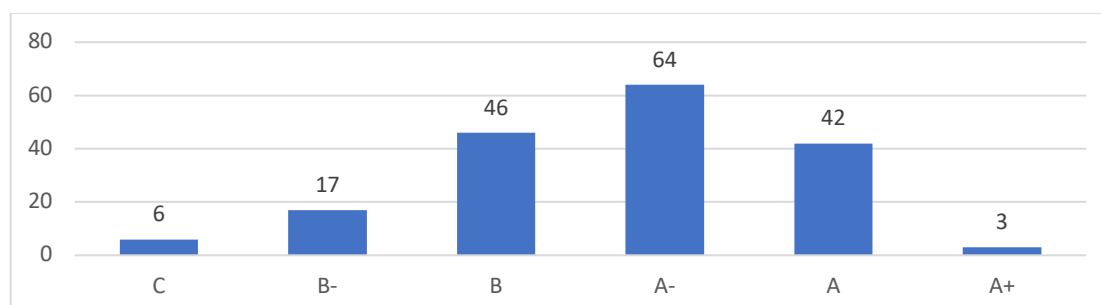
- **Online in the beginning - from (January 6th to January 23rd , on the scheduled slots)**
- **Offline during the month of February (January 26th to February 25th , on the scheduled days)**
- **Online for the remainder of the semester – on <https://iitgn-ac-in.zoom.us/j/9032601532>**

Assessment - Individual (open book) exam in the middle of the semester – Examination I period, and a group assignment by the end of the semester - assignment topics will depend on the students' profiles (engineering in general, computer science in particular, social sciences or humanities) and will be flexible

Challenges in the Digital Society
1. Introduction & Overview
2. Ethics
2.1 Ethical dimensions – framework and examples
2.2 Culture and Ethics – variation in the perspectives
2.3 Ethics of Artificial Intelligence
3. Privacy and Trust
3.1 Privacy basics – historical background and conceptual framework
3.2 Surveillance – levels and impacts
3.3 Privacy enhancing technologies
3.4 Cryptography basics – principles, techniques, and usage
4. Algorithms
4.1 Algorithm Bias – definitions and cases
4.2 Algorithm Transparency – concept and implications
5. Regulation and Law
5.1 Digital Regulations – Data Protection and AI
5.2. Compared Law between EU & India
6. Conclusions & Lessons

Some **bibliography** : Luciano Floridi (ed.) Information and Computer Ethics (2010) Cambridge University Press; Stahl B., Schroeder D. and Rodrigues, R., (2023) Ethics of Artificial Intelligence, Springer; Westin, A. (1967) Privacy and Freedom, Ig Publishing; Nissenbaum, H. (2010) Privacy in Context, Stanford University Press; Lyon, D. (2003) Surveillance as Social Sorting, Privacy and Discrimination, Routledge; Zuboff, S. (2019) The Age of Surveillance Capitalism, PublicAffairs; Adams, C. (2021) Introduction to Privacy Enhancing Technologies, Springer; Paar, C. (2010) Understanding cryptography a textbook for students and practitioners; Cobb, C. (2004) Cryptography For Dummies, Wiley Publishing; Baer, T. (2019) Understand, Manage, And Prevent Algorithmic Bias A Guide For Business Users And Data Scientists, Apress; Duke, T. (2023) Building Responsible AI Algorithms, Apress.

Previous figures (4 Editions, 178 students, grade distribution)



Based on

Challenges and Rights in the Digital Society	week
Bibliography / Reference Sources – ALWAYS UNDER REVISION	
1. Introduction & Overview	1
2. Ethics	
Royakkers, A., Timmer, J., Kool, L., Est, R. (2018) , Societal and ethical issues of digitization, Ethics and Information Technology	
Ethical aspects in Information Technologies – framework and examples	2
Luciano Floridi (ed.) Information and Computer Ethics (2010) Cambridge University Press, ISBN 13 978-0-521-71772-4	
Culture and Ethics – variation in perspectives	3
Hongladarom, S., and Ess C. (2007), Information Technology Ethics: Cultural Perspectives, Idea Group	
3. Privacy and Trust	
Privacy basics – historical background and conceptual framework	4
Westin, A. (1967) Privacy and Freedom, Ig Publishing	
Nissenbaum, H. (2010) Privacy in Context, Stanford University Press	
Surveillance – levels and impacts	5
Lyon, D. (2003) Surveillance as Social Sorting, Privacy and Discrimination, Routledge	
Zuboff, S. (2019) The Age of Surveillance Capitalism, PublicAffairs	
Privacy enhancing technologies	6
Adams, C. (2021) Introduction to Privacy Enhancing Technologies, Springer	
Cryptography basics – principles, techniques, and usage	7
Paar, C. (2010) Understanding cryptography a textbook for students and practitioners	
Cobb, C. (2004) Cryptography for Dummies Wiley Publishing	
Trust in the Digital Economy – needs and relevance	8
Ryan, P. (2019) Trust and Distrust in Digital Economies, Routledge	
4. Living with Algorithms	
Algorithm Bias – definitions and cases	9
Baer, T. (2019) Understand, Manage, And Prevent Algorithmic Bias A Guide For Business Users And Data Scientists, APRESS	
Algorithm Transparency – concept and implications	10
Lu, J., Lee, D., Kim, T.W. and Danks, D. (2019) Good Explanation for Algorithmic Transparency. SSRN: https://ssrn.com/abstract=3503603 or http://dx.doi.org/10.2139/ssrn.3503603	
5. Regulation and Law	
Compared Digital Rights Regulations – EU & India	11
Synodinou, T., Jogleux, P., Markou, C. and Pastitou, T. (eds) (2020) EU Internet Law in the Digital Era Regulation And Enforcement, Springer	
The Internet Normative Order – network norms	12
Kettermann, M (2020) The Normative Order of the Internet, Oxford UP	
6. Conclusions & Lessons	13